

Design Calculation Sheet - VBF-T5R1NOCEILING-CALC-01

PDF release of the xlsx deliverable; values mechanically taken from tool outputs (cell/*.json, parameter set).

Sheet: Inputs

Quantity	Value	Source
Positive electrode thickness	95.3 um	candidates/r4_A_params.json
Negative electrode thickness	107.5 um	candidates/r4_A_params.json
Separator thickness	8.0 um	candidates/r4_A_params.json
Positive current collector thickness (Al)	6.0 um	candidates/r4_A_params.json
Negative current collector thickness (Cu)	4.0 um	candidates/r4_A_params.json
Electrode width	2.37 m	candidates/r4_A_params.json
Electrode height	0.065 m	parameter set (Chen2020)
Electrode area	0.15405 m2	cell/r4_A_calc_dfn.json:area_m2
Positive / negative particle radius	1.5 / 1.0 um	candidates/r4_A_params.json
Electrolyte conductivity	2.0 S/m	candidates/r4_A_params.json (literature-advanced estimate)
Cation transference number	0.45	candidates/r4_A_params.json
Electrolyte diffusivity	4.50e-10 m2/s	candidates/r4_A_params.json (literature-advanced estimate)
Total heat transfer coefficient	200 W/m2/K	candidates/r4_A_params.json
Positive / negative porosity	0.335 / 0.25	parameter set (Chen2020)
Positive / negative density	3262.0 / 1657.0 kg/m3	parameter set (Chen2020)
Current collector densities (Al/Cu)	2700.0 / 8960.0 kg/m3	parameter set (Chen2020)
Voltage cut-offs	2.5 / 4.2 V	parameter set (Chen2020)
1C / 4C charge current	5 / 20 A	protocol definitions

Sheet: CapacityEnergy

Quantity	Value	Source
Capacity (1C to 2.5 V)	9.6251 Ah	cell/r4_A_calc_dfn.json:capacity_ah
Discharge energy	35.446 Wh	cell/r4_A_calc_dfn.json:energy_wh
Midpoint voltage	3.9016 V	cell/r4_A_calc_dfn.json:midpoint_voltage_v
DC resistance	0.001092 Ohm	cell/r4_A_calc_dfn.json:dcr_ohm
Power density	64930.1 W/kg	cell/r4_A_calc_dfn.json:power_density_w_kg
4C charge accepted (DFN)	8.12 Ah in 1461.0 s to 4.2 V	cell/r4_A_4c_dfn.json span x 20 A / 3600

Sheet: EnergyDensity

Quantity	Value	Source
Cell mass (contract caliber)	0.06070 kg	cell/r4_A_calc_dfn.json:mass_kg
Gravimetric energy density	583.93 Wh/kg	cell/r4_A_calc_dfn.json:energy_density_wh_kg
Volumetric energy density	1042.10 Wh/L	cell/r4_A_calc_dfn.json:energy_density_wh_l
Contract threshold	>= 500.94 Wh/kg	log.jsonl entry 0 (task text verbatim)
Positive / negative layer mass	31.846 / 20.580 g	calc layer_kg_m2 x area
Al / Cu collector mass	2.496 / 5.521 g	calc layer_kg_m2 x area
Separator mass	0.259 g	calc layer_kg_m2 x area

Sheet: NPandMass

Quantity	Value	Source
x_n charge window (2.5 -> 4.2 V)	0.0283 -> 0.7622	diag_r4a_np.py (DFN 4C experiment)
x_p charge window (2.5 -> 4.2 V)	0.8485 -> 0.3805	diag_r4a_np.py
Negative areal capacity	52.546 Ah/m2	diag_r4a_np.py (Q_n formula)
Positive areal capacity	50.161 Ah/m2	diag_r4a_np.py (Q_p formula)
N/P ratio	1.0475	diag_r4a_np.py (Q_n / Q_p)
Stack thickness	220.8 um	cell/r4_A_calc_dfn.json:thickness_m

Sheet: ProcessParams

Quantity	Value	Source
Positive / negative areal density	206.73 / 133.60 g/m ²	calc layer_kg_m2
Positive / negative compaction density	2.169 / 1.243 g/cm ³	parameter set: rho x (1-eps) / 1000
Electrolyte fill (volume / mass)	9.64 mL / 11.565 g	pore volume x 1.2 g/cm ³ (literature density), fill factor 1.0
Formation recommendation	0.1C CC to 4.2 V at 25 degC, 2 cycles	design recommended value (estimate)